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Joint time-vertex linear canonical transform

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ABSTRACT

The emergence of graph signal processing (GSP) has spurred a deep interest in signals that naturally reside on irregularly structured data kernels, such as those found in social, transportation, and sensor networks. Recently, concepts and applications related to time-varying graph signal analysis have matured, linking temporal signal processing techniques with innovative tools in GSP. In this paper, similar to the extension of the graph fractional Fourier transform to the graph linear canonical transform, we define the joint time-vertex linear canonical transform (JLCT) and its properties. This transformation extends the joint time-vertex Fourier transform (JFT) and fractional Fourier transform (JFRFT), broadening the Fourier analysis in both time and vertex domains into the domain of the linear canonical transform (LCT). This offers an enhanced set of the LCT analysis tools for joint time-vertex processing. Applications of the JLCT in dynamic mesh denoising, clustering, and energy compactness demonstrate that JLCT can enhance regression and learning tasks, and can refine and improve the performance of the JFT and the JFRFT.

1. Introduction

1.1. Graph signal processing overview

In traditional fields such as communication systems, radar applications, and medical imaging, classical discrete signal processing (DSP) based on Fourier analysis has been the cornerstone framework for handling digital signals. It is evident that classical DSP tools are well-suited for discrete signals within uniform, regular grids. However, with the increasing volume of complex data in non-Euclidean spaces and irregular domains, such as social networks, transportation systems, sensor networks, and biological connectomes, the DSP cannot be directly applied to these signals.

Consequently, the theory and applications of the graph signal processing (GSP) [1–6] have rapidly evolved in recent years. The GSP defines its network elements as graph vertices and their interrelated edges. Signals are associated with the vertices of graphs, thus defining graph signals. To process and analyze these graph signals, many have extended classical DSP tools to GSP, including graph transforms [1–4], frequency analysis [7,8], filtering [9,10], sampling [11–15], reconstruction [16,17], the graph uncertainty principle [12,18], fast algorithms [13,19], and deep learning [20], among others. In the GSP framework, two standard signal processing approaches have been adopted [1,2]. The first, inspired by harmonic analysis [1], utilizes the graph Laplacian operator as its core theoretical foundation. The second, inspired by algebraic methods [2], uses the basic shift operator of the underlying graph, namely the adjacency matrix, as the main computational core. These two approaches provide complementary perspectives and tools, collectively fostering attention and development in the inclusive field [6].

1.2. Time-vertex signal processing

Traditional GSP theory relies on the transformation of orthogonal bases within finite-dimensional vector spaces [1–4]. In this context, a graph signal f assigns a real number to each vertex, denoted as $f \in \mathbb{R}^N$, where $\mathbb{R}$ represents the set of real numbers. However, each vertex can be assigned not only a real number but also temporal information [21]. Specifically, when a vertex corresponds to a specified closed interval $[a, b]$ following the Lebesgue measure, it is recognized within the framework of joint time-vertex GSP [22]. In recent years, concepts and applications related to the joint harmonic analysis of time-varying graph signals have matured. For instance, the joint time-vertex Fourier transform (JFT) and associated filtering techniques link time-domain signal pro-

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cessing methodologies with new tools from GSP [23–25]. These pivotal technologies, known as time-vertex GSP, integrate the discrete Fourier transform (DFT) from the time domain with the graph Fourier transform (GFT) from the vertex domain. Time-vertex GSP has been deployed in various applications, including the reconstruction of time-varying graph signals [26–28], prediction of joint spectral-time data [29], and forecasting the evolution of static graph signals [30]. Additionally, transformations and filtering in spatio-temporal graphs [31,32], semi-supervised learning [33,34] have been explored.

While the JFT effectively facilitates basic graph frequency analysis within its framework, it operates similarly to the DFT and GFT. The JFT transforms purely from the time domain to the frequency domain, and from the vertex domain to the graph spectral domain, respectively. These transformations rely on the entirety of the graph signal across the time-vertex domain, but lack flexibility during the conversion process from the time-vertex domain to the graph frequency domain. Due to these constraints, the application of the JFT across different domains remains quite limited. To address these issues, Koç et al. [35] combined the discrete fractional Fourier transform (DFRFT) [36] and the graph fractional Fourier transform (GFRFT) [37] to propose the joint time-vertex fractional Fourier transform (JFRFT), an extension of the JFT. The JFRFT introduces two parameters, allowing for the analysis of signals within both the fractional time domain and the fractional graph spectral domain, thereby providing enhanced observational flexibility through its dual-domain nature. Despite the significant contributions of these techniques, we believe that the potential of incorporating classical DSP transformation methods into joint harmonic analysis has not been fully exploited, especially when dealing with non-stationary signals and fractional-order scenarios. Additionally, current methods still lack the capability to perform more complex operations such as scaling, shearing, and frequency modulation. These limitations have spurred the need for novel, more versatile approaches to time-vertex domain processing of graph signals within an inclusive framework.

1.3. Linear canonical transform

To address the issues outlined above, we have applied the linear canonical transform (LCT) [38] to GSP. Originally introduced in the 1970s and later adopted in signal processing [39] in 2001, the LCT generalizes several transformations including the Fourier transform (FT), fractional Fourier transform (FRFT), Fresnel transform, and scaling operations, characterized by three free parameters. The LCT of a signal $x(t)$ with parameters (a, b, c, d) is defined as [40]

$$\mathcal{O}_{LCT}^{\mathbf{K}}(x(t)) = \begin{cases} \sqrt{\frac{1}{ib}} \cdot \int_{-\infty}^{\infty} \exp\left[i\pi\left(\frac{d}{b}u^2 - \frac{2}{b}ut + \frac{a}{b}t^2\right)\right] x(t)\,\mathrm{d}t & \text{if } b \neq 0, \\ \sqrt{d} \cdot \exp\left(i\pi dct^2\right) x(d \cdot t) & \text{if } b = 0, \end{cases} \tag{1}$$

where $\mathcal{O}_{LCT}^{\mathbf{K}}$ denotes the LCT operator and the matrix $\mathbf{K}$ has entries $[a, b; c, d]$, known as the LCT parameter matrix, with the requirement that $ad - bc = 1$ and $a, b, c, d \in \mathbb{R}$.

In recent years, the application of the discrete linear canonical transform (DLCT) has become increasingly widespread. Several advanced variants have been proposed, including the DLCT on the hyperdifferential operator [41], the sliding DLCT [42], and the sparse DLCT [43]. Additionally, quaternions and octonions have been incorporated into the framework of DLCT [44,45].

1.4. Main contributions

In our previous research [46], we extended the LCT to graphs, introducing the graph linear canonical transform (GLCT). As a generalization of the GFT and the GFRFT, the GLCT demonstrates strong adaptability and flexibility in signal processing due to its three free parameters. The GLCT offers versatility in shaping spectral characteristics, allowing for signal compression, expansion, and rotation within the vertex frequency plane to form various smoothness patterns. We have demonstrated that it satisfies all the expected properties, unifying GFT, GFRFT, and graph-scale transformations. Parallel to the extension of the JFT through the JFRFT, we introduce the joint time-vertex linear canonical transform (JLCT), which enables the analysis of signals within the time domain under LCT and within the GLCT domain. Integrating the DLCT [40] in the time domain with the GLCT in the vertex domain within the joint time-vertex GSP framework not only capitalizes on the strengths of DLCT in handling time series data but also leverages GSP to globally process time-varying intervals and vertices jointly, effectively utilizing GLCT.

The JLCT embodies the characteristics of both the DLCT and the GLCT, offering significant flexibility and versatility. In the analysis and processing of non-stationary signals, the JLCT enables focused attention on both temporal and vertex-specific local features as well as frequency characteristics by adjusting parameters, and can implement operations such as scaling, cutting, and frequency modulation, thereby providing an extensive range of time-frequency analysis capabilities. We present and verify several properties of the JLCT, such as index additivity, invertibility, separability, reducibility, identity, and unitary. To advance this theory further, we introduce a foundational denoising theory based on localized filters within the proposed JLCT framework. We apply the JLCT to dynamic mesh denoising, clustering, and energy compactness comparisons, and experimental results demonstrate its enhancement of regression and learning tasks, exhibiting superior performance compared to the JFT and the JFRFT.

We summarize the novelties and contributions as follows.

- The JLCT is introduced as a novel approach for analyzing and processing time-vertex signals within a broader category of transformations.
- In the JLCT framework, a foundational theoretical setup for denoising based on the localized filter is proposed.
- Denoising, clustering, and energy compaction comparisons were conducted using JLCT, demonstrating its enhancement of regression and learning tasks, as well as its superior performance.

1.5. Outline

The remainder of this paper is organized as follows. In Section 2, we provide preliminary information on the DLCT, GLCT and joint time-vertex processing. In Section 3, we define the proposed JLCT and provide its properties. In Section 4, the theory of denoising based on the localized filter is presented, and three numerical experiments demonstrate performance of the JLCT in reconstruction, denoising, and clustering. In Section 5, we conclude the paper.

2. Preliminaries

In this section, we provide a concise overview of the fundamental theoretical concepts concerning DLCT based on dilated Hermite functions [40] and graph signals [1–4], encompassing the GLCT [46] as well as the JFT and JFRFT [22,35].

2.1. Discrete linear canonical transform

For discrete signals $x[n]$, the DLCT can be implemented using a method that adopts the center discrete dilated Hermite functions (CDDHFs) eigendecomposition [40], expressed as follows

$$\mathcal{O}_{DLCT}^{\mathbf{K}}(x[n]) = \mathbf{O}_D^{\mathbf{K}} \cdot x[n] = \left(\mathbf{D}^{\xi}\mathbf{E}_{\sigma}\mathbf{D}^{\alpha}\mathbf{E}^{-1}\right) x[n], \tag{2}$$

where $\mathcal{O}_{DLCT}^{\mathbf{K}}$ and $\mathbf{O}_D^{\mathbf{K}}$ refer to the DLCT operator and the DLCT matrix, respectively, and parameter matrix $\mathbf{K}$ is the same as Eq. (1). Furthermore, due to the unitarity of $\mathbf{E}$, we have $\mathbf{E}^{-1} = \mathbf{E}^{\top}$.

The matrix operations in Eq. (2), are delineated through a three-stage process, where the indices satisfy $0 \le n, m \le N - 1$:

1) *DFRFT Stage*: Due to the non-commutativity of the LCT matrix, the DFRFT stage is implemented first, with its parameter matrix $[\cos\alpha, \sin\alpha; -\sin\alpha, \cos\alpha]$, and is captured by $\mathbf{D}^{\alpha} = \mathrm{diag}\{\exp[-i\frac{\pi}{2}(m + \frac{1}{2})\alpha]\}$, $\alpha \in [0,1]$, where the range of α is the range given in [40], divided by $\pi/2$.
2) *Scaling Stage*: The discrete scaling stage is realized using CDDHFs as the expanded eigenspace. The matrix $\mathbf{E}_{\sigma}$ is composed of the eigenvectors of the matrix $\mathbf{H}_{\sigma}$, where $\mathbf{H}_{\sigma} = \mathbf{T} + \sigma^4\mathbf{F}\mathbf{T}\mathbf{F}^{-1}$, with $\mathbf{T}$ being a weighted diagonal matrix and $\mathbf{F}$ representing the central DFT matrix [47]. $\mathbf{E}^{-1}$ denotes the inverse of the matrix $\mathbf{E}_{\sigma}$ when $\sigma = 1$.
3) *Chirp Multiplication (CM) Stage*: The CM stage is expressed as a diagonal matrix with chirp function elements $\mathbf{D}^{\xi} = \mathrm{diag}\{\exp[i\pi\xi(n - \frac{N-1}{2})^2/N]\}$.

The relationships between the parameter matrix $\mathbf{K}$ and (ξ, σ, α) are defined as

$$\xi = \frac{ac+bd}{a^2+b^2}, \quad \sigma = \sqrt{a^2+b^2}, \quad \alpha = \frac{2}{\pi}\cos^{-1}\left(\frac{a}{\sigma}\right) = \frac{2}{\pi}\sin^{-1}\left(\frac{b}{\sigma}\right). \tag{3}$$

2.2. Graph signals and graph linear canonical transform

For arbitrary weighted graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$, where $\mathcal{V} = \{v_0, v_1, \ldots, v_{N-1}\}$ represents the set of vertices with size $|\mathcal{V}| = N$, and $\mathcal{E}$ denotes the set of edges. A graph signal $\mathbf{x} \in \mathbb{C}^N$ is defined as a mapping $\mathbf{x} : \mathcal{V} \to \mathbb{C}^N$, i.e., $v_n \to \mathbf{x}_n$. The adjacency matrix $\mathbf{A} \in \mathbb{R}^{N\times N}$ captures the weight on each edge and delineates the pairwise relationships among vertices of the graph. For undirected graphs, $\mathbf{A}$ is symmetric. The adjacency matrix can define the graph Laplacian matrix, and its eigenvectors can also utilized to define the GFT [2,4].

Based on the definition of the DLCT, we can analogously introduce the concept of the GLCT [46]. Initially, the GFT [1–4], which utilizes the eigendecomposition of the adjacency matrix $\mathbf{A} = \mathbf{V}\mathbf{\Lambda}_{\mathbf{A}}\mathbf{V}^{-1}$ is defined as

$$\hat{\mathbf{x}} = \mathbf{F}_{\mathcal{G}}\mathbf{x} = \mathbf{V}^{-1}\mathbf{x} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{-1}\mathbf{x}, \tag{4}$$

where the columns of $\mathbf{V}$ are the eigenvectors of $\mathbf{A}$, and the matrices $\mathbf{Q}$ and $\mathbf{\Lambda}$ are given by the spectral decomposition of the GFT matrix $\mathbf{F}_{\mathcal{G}}$, which is orthogonal and diagonalized.

Definition 1. Parallel to the GFT, the GLCT of $\mathbf{x} \in \mathbb{C}^N$ can be defined as

$$\hat{\mathbf{x}} = \mathcal{O}^{\mathbf{M}}_{GLCT}(\mathbf{x}) = \mathbf{O}^{\mathbf{M}}_{\mathcal{G}} \cdot \mathbf{x} = \left(\mathbf{\Lambda}^{\epsilon}\mathbf{Q}_{\varsigma}\mathbf{\Lambda}^{\beta}\mathbf{Q}^{-1}\right)\mathbf{x}, \tag{5}$$

where the operator $\mathbf{O}^{\mathbf{M}}_{\mathcal{G}}$ signifies a GLCT operator matrix, with the matrix $\mathbf{M}$ characterized by entries $[e, f; g, h]$, known as the GLCT parameter matrix. This matrix adheres to the condition $eh - fg = 1$, with $e, f, g, h \in \mathbb{R}$.

The matrix operations for the GLCT can also be divided into three stages:

1) *GFRFT Stage*: By conducting an eigendecomposition of the orthogonal and diagonalized GFT matrix $\mathbf{F}_{\mathcal{G}}$ [37], the GFRFT matrix is defined as $\mathbf{\Lambda}^{\beta} = \mathrm{diag}(\lambda_i^{\beta})$ for $i = 0, \ldots, N-1$, where λ_i are the eigenvalues of $\mathbf{F}_{\mathcal{G}}$.
2) *Graph Scaling Stage*: By performing eigendecomposition on the scaled adjacency matrix $\varsigma \cdot \mathbf{A}$ [9], the scaled GFT matrix $(\mathbf{F}_{\mathcal{G}})_{\varsigma}$ is obtained. Further eigendecomposition results in a matrix $\mathbf{Q}_{\varsigma}$ composed of the eigenvectors of $(\mathbf{F}_{\mathcal{G}})_{\varsigma}$.
3) *Graph CM Stage*: The CM matrix is given by $\mathbf{\Lambda}^{\epsilon} = \mathrm{diag}(\exp(j\frac{\pi}{2}k\epsilon))$, where $\epsilon = lk + f$ represents the CM parameter and is a linear function of k with constants l and f.

Furthermore, its inverse GLCT (IGLCT) is

$$\mathbf{x} = (\mathbf{O}^{\mathbf{M}}_{\mathcal{G}})^{-1} \cdot \hat{\mathbf{x}} = \mathbf{O}^{\mathrm{inv}(\mathbf{M})}_{\mathcal{G}} \cdot \hat{\mathbf{x}} = \left(\mathbf{Q}\mathbf{\Lambda}^{-\beta}\mathbf{Q}_{\varsigma}^{-1}\mathbf{\Lambda}^{-\epsilon}\right)\hat{\mathbf{x}}, \tag{6}$$

The parameter relationship between $\mathbf{M}$ and $(\epsilon, \varsigma, \beta)$ is consistent with Eq. (3).

Remark 1. In particular, when $e = \cos\beta$, $f = \sin\beta$, $g = -\sin\beta$, and $h = \cos\beta$, the GLCT corresponds to the GFRFT [37]. Conversely, when $e = 0$, $f = 1$, $g = -1$, and $h = 0$, the GLCT reduces to the GFT.

Remark 2. Since the GLCT is generalized from the DLCT in cyclic graphs, and the Laplacian operator cannot be applied to directed graphs, we choose the adjacency matrix within the algebraic method framework to broaden its applicability.

2.3. Joint time-vertex signal processing

For the purpose of joint time-vertex processing, it becomes imperative to formalize the concept of a "time-vertex signal". Specifically, let $\mathbf{X}$ and its vectorized form $\mathbf{x} = \mathrm{vec}(\mathbf{X}) \in \mathbb{C}^{NT}$ denote *time-vertex signals*. Assume that the signals on the graph are sampled at regular continuous intervals of unit length T [22]. If we use the vector $\mathbf{x}_t \in \mathbb{C}^N$ to represent the graph signal at time t, then the matrix corresponding to the time-varying graph signal is $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T] \in \mathbb{C}^{N\times T}$. We denote $\mathbf{X}^{\top}$, $\mathbf{X}^*$, and $\mathbf{X}^{\mathrm{H}}$ as the transpose, complex conjugate, and Hermitian (conjugate transpose) of the matrix $\mathbf{X}$, respectively.

2.3.1. Joint time-vertex Fourier transform

For the signal $\mathbf{X}$, which contains graph signals in its columns and defines time series signals for each vertex of the base graph in its rows, the DFT of this signal is given by

$$\mathrm{DFT}\{\mathbf{X}\} = \mathbf{X}\mathbf{F}_{\mathcal{D}}^{\top}, \tag{7}$$

where $\mathbf{F}_{\mathcal{D}}$ is the normalized DFT matrix [48]. The elements of $\mathbf{F}_{\mathcal{D}}$ are defined as $\exp(-j2\pi tk/T), t, k = 0, 1, \ldots, T-1$. While the GFT of $\mathbf{X}$ can be given as

$$\mathrm{GFT}\{\mathbf{X}\} = \mathbf{F}_{\mathcal{G}}\mathbf{X}, \tag{8}$$

where $\mathbf{F}_{\mathcal{G}}$ is defined in Eq. (4). Therefore, based on Eqs. (7) and (8), the JFT of $\mathbf{X}$ can be defined as

$$\mathrm{JFT}\{\mathbf{X}\} = \mathbf{F}_{\mathcal{G}}\mathbf{X}\mathbf{F}_{\mathcal{D}}^{\top}, \tag{9}$$

which is capable of capturing the spectral information of $\mathbf{X}$ both through the lens of time and from the perspective of the underlying graph. In general, graph signals are often represented in vector form. Therefore, the transformation becomes

$$\hat{\mathbf{x}} = \mathrm{JFT}\{\mathbf{x}\} = \mathbf{J}\mathbf{x} = \left(\mathbf{F}_{\mathcal{D}} \otimes \mathbf{F}_{\mathcal{G}}\right)\mathbf{x}, \tag{10}$$

where $\otimes$ is the Kronecker product, and $\mathbf{x} = \mathrm{vec}(\mathbf{X})$. The relationship between Eqs. (9) and (10) is derived through the properties of the Kronecker product.

2.3.2. Joint time-vertex fractional Fourier transform

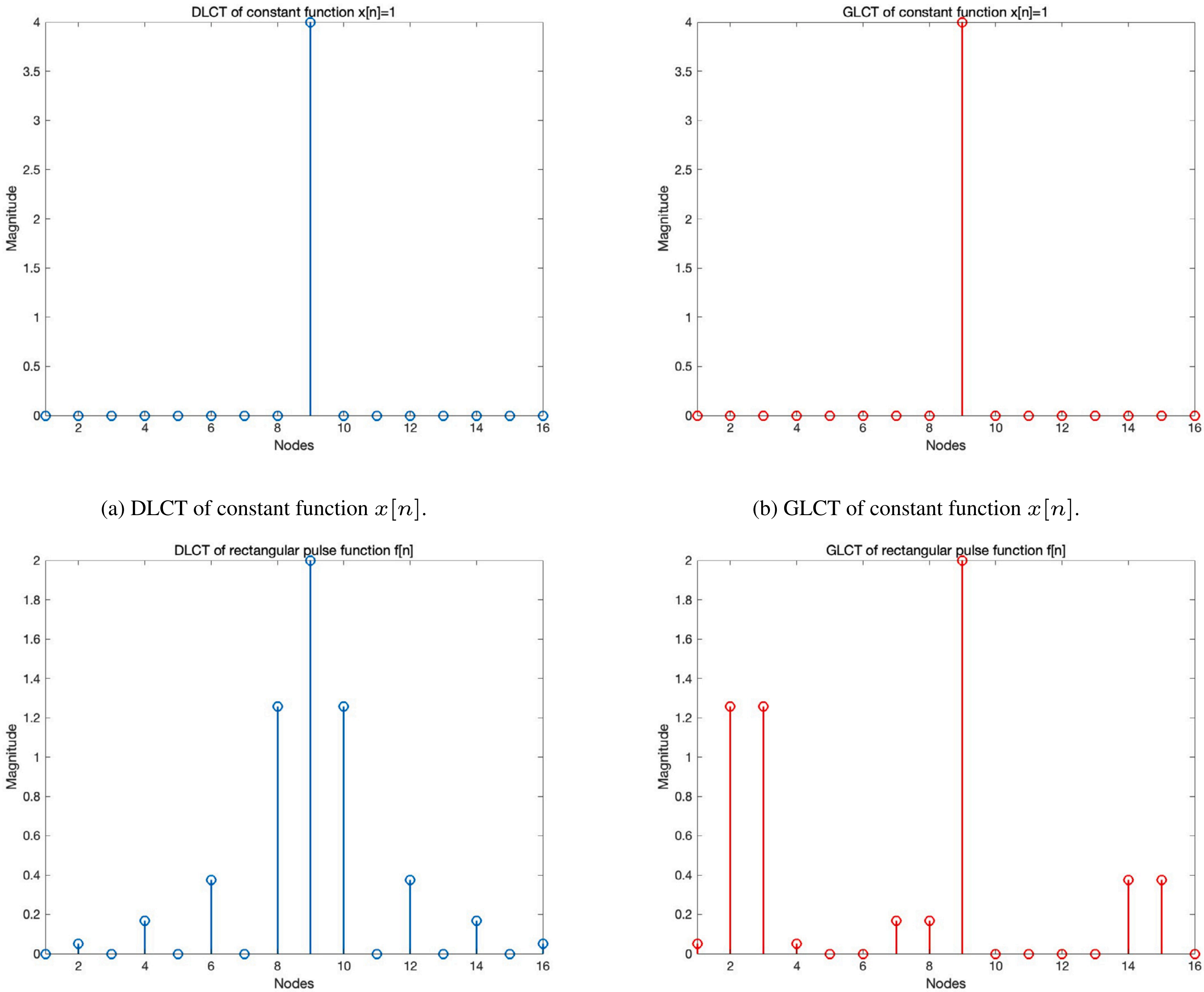
Similar to the JFT, a JFRFT can be defined in the fractional domain, extending the concept to both the DFRFT domain and the GFRFT domain. For a pair of orders (α, β), where $\alpha, \beta \in [0,1]$, the JFRFT of a graph signal $\mathbf{X} \in \mathbb{C}^{N\times T}$ can be defined as

$$\hat{\mathbf{X}} = \mathrm{JFRFT}\{\mathbf{X}\} = \mathbf{F}_{\mathcal{G}}^{\beta}\mathbf{X}(\mathbf{F}_{\mathcal{D}}^{\alpha})^{\mathrm{H}}, \tag{11}$$

where the βth order GFRFT is defined as $\mathbf{F}_{\mathcal{G}}^{\beta} = \mathbf{Q}\mathbf{\Lambda}^{\beta}\mathbf{Q}^{-1}$, which is the same as the *GFRFT stage*, and $\mathbf{F}_{\mathcal{D}}^{\alpha}$ denotes the αth order DFRFT [36].

When vectorizing a given time-vertex signal $\mathbf{X}$ into $\mathbf{x} = \mathrm{vec}(\mathbf{X})$, an equivalent transformation in the vectorized form can be established as

$$\hat{\mathbf{x}} = \mathrm{JFRFT}\{\mathbf{x}\} = \mathbf{J}^{\alpha,\beta}\mathbf{x} = \left(\mathbf{F}_{\mathcal{D}}^{\alpha} \otimes \mathbf{F}_{\mathcal{G}}^{\beta}\right)\mathbf{x}, \tag{12}$$



(a) DLCT of constant function $x[n]$.

(b) GLCT of constant function $x[n]$.

(c) DLCT of rectangular pulse function $f[n]$.

(d) GLCT of rectangular pulse function $f[n]$.

Fig. 1. Spectrum after DLCT and GLCT of the signal.

where $\mathbf{J}^{\alpha,\beta}(\cdot) : \mathbb{C}^{NT} \to \mathbb{C}^{NT}$ represents the JFRFT in its vectorized form, and the derivation of the equality on the right side is similar to Eq. (10).

3. Joint time-vertex linear canonical transform

To propose a novel transformation framework that integrates the concept of the LCT into temporal vertex signal processing, akin to the synthesis of the FT and FRFT, we introduce a framework termed the JLCT. The aim of establishing this framework is to delineate a frequency domain representation for temporal vertex signals in the GLCT domain, thereby incorporating additional parameters to enhance flexibility and facilitate exploration between the temporal vertex and spectral domains. In this section, we define the JLCT as a unified extension of the JFT and JFRFT into the DLCT and GLCT domains, respectively.

However, despite the GLCT being derived based on [40] in [46], there is a lack of correspondence with the GLCT due to the limitations of the DLCT based on CDDHFs. Although GLCT and DLCT are formally similar, the GLCT in a cyclic graph is not effectively equivalent to the DLCT under certain parameter configurations. Consequently, we propose a redefinition of a novel type of DLCT specifically for cyclic graphs, based on the CDDHF framework. The DLCT of this signal is given by

$$\mathrm{DLCT}\{\mathbf{X}\} = \mathbf{X}\left(\mathbf{O}_D^{\mathbf{K}}\right)^{\mathrm{H}} = \mathbf{X}\left(\mathbf{D}^{\xi}\mathbf{E}_{\sigma}\mathbf{D}^{\alpha}\mathbf{E}^{-1}\right)^{\mathrm{H}}, \tag{13}$$

where $\mathbf{D}^{\alpha} = \mathrm{diag}\{\exp[-i\pi m\alpha/2]\}$, $\mathbf{E}_{\sigma}$ denotes the eigenvector matrix corresponding to Hermite-Gaussian functions obtained after a new scaling transformation post-DFRFT [36], and the matrix for the CM stage $\mathbf{D}^{\xi} = \mathrm{diag}\{\exp[i\pi\xi(n-\frac{N}{2})^2/N]\}, 0 \le n, m \le T-1$. Appendix A provides a detailed derivation process. Moreover, to enhance compatibility and performance under cyclic graph conditions with the DLCT, we have modified the definitions of GLCT parameters ς and ϵ, thus enabling a refined definition of the JLCT.

A simple example, when the GLCT (with $\varsigma = 1$) and the DLCT correspond in the cycle graph, $n = 16$, as illustrated in Fig. 1, panels (a) and (b) display the spectra of the DLCT and GLCT for a constant signal $\mathbf{x}[n] = [1\ 1\ \ldots\ 1\ 1]^{\top}$ with the same parameter matrix $[0, 1; -1, 1/2]$, demonstrating identical outcomes. Panels (c) and (d) present the spectra for a rectangular pulse function signal $f[n]$ under the same parameter

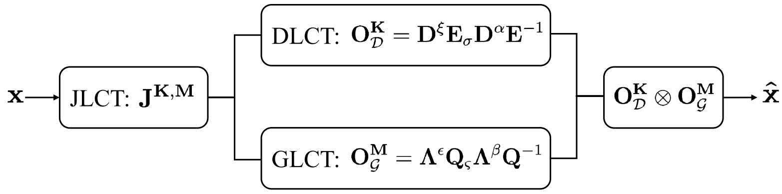


Fig. 2. Flowchart of the JLCT process.

matrix $[0, -1; 1, -2]$. Although the index positions differ, the magnitudes remain the same.

3.1. Definition

Prior to delving into the intricacies of the JLCT, it is imperative to understand the relationships among $\mathbb{C}^T$, $\mathbb{C}^N$, and $\mathbb{C}^{NT}$, as well as the inner products they carry. The tensor product $\mathbb{C}^T \otimes \mathbb{C}^N$ is endowed with an inner product $\langle \cdot, \cdot \rangle_{\mathbb{C}^T \otimes \mathbb{C}^N}$ that is linearly induced by

$$\langle t_1 \otimes v_1, t_2 \otimes v_2 \rangle_{\mathbb{C}^T \otimes \mathbb{C}^N} = \langle t_1, t_2 \rangle_{\mathbb{C}^T} \cdot \langle v_1, v_2 \rangle_{\mathbb{C}^N}, \tag{14}$$

thereby defining a metric on $\mathbb{C}^T \otimes \mathbb{C}^N$. Given that $\mathbb{C}^T$ and $\mathbb{C}^N$ are finite-dimensional, $\mathbb{C}^T \otimes \mathbb{C}^N$ is complete. Let $\Psi = \{\psi_i\}_{1 \le i \le T}$ be an orthonormal basis of $\mathbb{C}^T$, and let $\Phi = \{\phi_j\}_{1 \le j \le N}$ be an orthonormal basis of $\mathbb{C}^N$. Consequently, $\Psi \otimes \Phi = \{\psi_i \otimes \phi_j\}_{1 \le i \le T, 1 \le j \le N}$ forms an orthonormal basis of $\Psi \otimes \Phi$.

Employing the tensor product construction, we obtain an alternate characterization of $\mathbb{C}^{NT}$, demonstrating that it is a separable vector space.

Lemma 1. *$\mathbb{C}^{NT}$ constitutes a vector space, and there exists an isomorphism between $\mathbb{C}^{NT}$ and $\mathbb{C}^T \otimes \mathbb{C}^N$, represented by [49]*

$$\varphi(\mathbf{x}) = \sum_{v \in \mathcal{V}} \mathbf{x}(v) \otimes v, \quad \text{for } \mathbf{x} \in \mathbb{C}^{NT}, \tag{15}$$

where $\mathcal{V}$ corresponds to the standard basis of $\mathbb{C}^N$.

Proof. For $\mathbf{x}, \mathbf{y} \in \mathbb{C}^{NT}$, elements $v \in \mathcal{V}$, and a scalar $a \in \mathbb{C}$, the operations $(\mathbf{x}+\mathbf{y})(v) = \mathbf{x}(v) + \mathbf{y}(v) \in \mathbb{C}^T$ and $(a\mathbf{x})(u) = a\mathbf{x}(u)$ hold. This demonstrates that φ acts as an injection from $\mathbb{C}^{NT}$ to $\mathbb{C}^T \otimes \mathbb{C}^N$. Each element $b \in \mathbb{C}^T \otimes \mathbb{C}^N$ can be represented as $b = \sum_{v \in V} t_v \otimes v$, where $t_v \in \mathbb{C}^T$. Defining $\varphi^{-1}(b)(v) = t_v$ ensures that $\varphi^{-1}(b) \in \mathbb{C}^{NT}$ serves as the inverse mapping of φ. Thus, φ constitutes an isomorphism. □

Let $\mathbf{X} \in \mathbb{C}^{N \times T}$ represent a signal stored over a time sequence defined on a graph $\mathcal{G}$, where N is the number of vertices in $\mathcal{G}$, and T is the dimension of the signal in the time series defined at each vertex. We then define the parameter matrix for the DLCT as $\mathbf{K} = [a, b; c, d]$, and for the GLCT as $\mathbf{M} = [e, f; g, h]$. Consequently, the JLCT for the parameter matrix pair $(\mathbf{K}, \mathbf{M})$ is defined as follows

$$\hat{\mathbf{X}} = \text{JLCT}\{\mathbf{X}\} := \mathbf{O}_{\mathcal{G}}^{\mathbf{M}} \mathbf{X} \left(\mathbf{O}_{\mathcal{D}}^{\mathbf{K}}\right)^{\mathrm{H}}, \tag{16}$$

where $\mathbf{O}_{\mathcal{D}}^{\mathbf{K}}$ is the DLCT operator defined in the time domain corresponding to the parameter matrix $\mathbf{K}$ as specified in Eq. (13). Concurrently, $\mathbf{O}_{\mathcal{G}}^{\mathbf{M}}$ is the GLCT operator for the parameter matrix $\mathbf{M}$ defined in Eq. (5).

Remark 3. To enhance compatibility and performance with the DLCT while preserving the essential definitions, we have modified the interpretation of parameters ς and ϵ. Specifically, during the graph scaling stage, we define the form: $\varsigma^4\mathbf{A}$. For the graph CM stage, defined as $\mathbf{\Lambda}^{\epsilon} = \text{diag}(\lambda^{\epsilon}) = \text{diag}\{\exp[j\pi/2 \cdot k(2 - 2k/N - N/2)l]\}$, where $\epsilon = (2 - 2k/N - N/2)l$ encapsulates the information of chirp function elements and is singularly dependent on the variable l.

Similar to the construction of the JFT and the JFRFT, by vectorizing the given time-vertex signal $\mathbf{X}$ into $\mathbf{x} = \text{vec}(\mathbf{X})$, we can identify that the equivalent transformation between the vectorized forms also maps from $\mathbb{C}^{NT}$ to $\mathbb{C}^{NT}$.

Definition 2. For $\mathbf{x} \in \mathbb{C}^{NT}$, the JLCT of $\mathbf{x}$ is defined as

$$\hat{\mathbf{x}} = \mathbf{J}^{\mathbf{K},\mathbf{M}}\mathbf{x} = \left(\mathbf{O}_{\mathcal{D}}^{\mathbf{K}} \otimes \mathbf{O}_{\mathcal{G}}^{\mathbf{M}}\right)\mathbf{x}, \tag{17}$$

where $\otimes$ denotes Kronecker product, and

$$\left(\mathbf{O}_{\mathcal{D}}^{\mathbf{K}} \otimes \mathbf{O}_{\mathcal{G}}^{\mathbf{M}}\right)\mathbf{x} = \text{vec}(\mathbf{O}_{\mathcal{G}}^{\mathbf{M}}\mathbf{X}\left(\mathbf{O}_{\mathcal{D}}^{\mathbf{K}}\right)^{\mathrm{H}}).$$

Similar to **Remark** 1, the JLCT operator reduces to the JFRFT or JFT operator when its parameters take specific values. Specifically, when $\xi = \epsilon = 0$ and $\sigma = \varsigma = 1$, the JLCT operator becomes the JFRFT operator [35]. Further, when $\xi = \epsilon = 0$, $\sigma = \varsigma = 1$, and $\alpha = \beta = 1$, the JLCT operator reduces to the JFT operator. This is explained in detail in **Property** 5. We have depicted the process of the JLCT more clearly using the flowchart in Fig. 2.

We can form an orthonormal basis Ψ for $\mathbb{C}^T$ with the eigenvector matrix $(\mathbf{O}_{\mathcal{D}}^{\mathbf{K}})^{-1}$, and Φ as an orthonormal basis for $\mathbb{C}^N$, comprising the eigenvector matrix $(\mathbf{O}_{\mathcal{G}}^{\mathbf{M}})^{-1}$. Here $\Psi = [\psi_0, \ldots, \psi_{T-1}] \in \mathbb{C}^{T \times T}$ with $\psi_t = [\psi_{t,0}, \ldots, \psi_{t,T-1}]^{\top} \in \mathbb{C}^T$ for $t \in \{0, \ldots, T-1\}$, and $\Phi = [\phi_0, \ldots, \phi_{N-1}] \in \mathbb{C}^{N \times N}$ with $\phi_n = [\phi_{n,0}, \ldots, \phi_{n,N-1}]^{\top} \in \mathbb{C}^N$ for $n \in \{0, \ldots, N-1\}$.

When representing the JLCT using the inner product of vectors and standard orthonormal bases, we introduce the operator

$$\mathcal{O}_{\mathbf{x}}^{\mathbf{K},\mathbf{M}}(\psi \otimes \phi) = \langle \mathbf{x}, \psi \otimes \phi \rangle := \langle \varphi(\mathbf{x}), \psi \otimes \phi \rangle_{\mathbb{C}^T \otimes \mathbb{C}^N}, \tag{18}$$

where, φ denotes the isomorphism mapping as detailed in Lemma 1, and the inner product on the right side of Eq. (18) is defined in accordance with Eq. (14).

The JLCT of $\mathbf{x}$ can be decomposed into simpler components, referred to as partial JLCTs, which are represented as operators

$$\mathcal{O}_{\mathbf{x}}^{\mathbf{K}}(\psi)(v) = \langle \mathbf{x}(\cdot, v), \psi \rangle_{\mathbb{C}^T} = \left(\mathbf{O}_{\mathcal{D}}^{\mathbf{K}} \otimes \mathbf{I}_N\right)\mathbf{x}, \tag{19}$$

and

$$\mathcal{O}_{\mathbf{x}}^{\mathbf{M}}(\phi)(t) = \langle \mathbf{x}(t, \cdot), \phi \rangle_{\mathbb{C}^N} = \left(\mathbf{I}_T \otimes \mathbf{O}_{\mathcal{G}}^{\mathbf{M}}\right)\mathbf{x}, \tag{20}$$

for every $v \in \mathcal{V}$ and $t \in \mathbb{C}^T$, and the matrices $\mathbf{I}_N$ and $\mathbf{I}_T$ are the identity matrices for dimensions N and T, respectively.

3.2. Properties

In this section, we delineate the key properties, propositions, and corollaries of the introduced JLCT framework, accompanied by their respective proofs.

Property 1. ***Additivity:*** *The additivity of the LCT is expressed through matrix multiplication, and the additivity of the JLCT can be described by substituting matrices with*

$$\mathbf{J}^{\mathbf{K}_3,\mathbf{M}_3} = \mathbf{J}^{\mathbf{K}_1\mathbf{K}_2,\mathbf{M}_1\mathbf{M}_2} = \mathbf{J}^{\mathbf{K}_1,\mathbf{M}_1} \cdot \mathbf{J}^{\mathbf{K}_2,\mathbf{M}_2}, \tag{21}$$

where $(\mathbf{K}_1,\mathbf{M}_1)$*,* $(\mathbf{K}_2,\mathbf{M}_2)$*, and* $(\mathbf{K}_3,\mathbf{M}_3)$ *represent pairs of parameter matrices, and* $\mathbf{K}_3=\mathbf{K}_1\mathbf{K}_2, \mathbf{M}_3=\mathbf{M}_1\mathbf{M}_2$*, i.e.*

$$\begin{bmatrix} a_3 & b_3 \\ c_3 & d_3 \end{bmatrix} = \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix} \cdot \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix}, \text{and}$$
$$\begin{bmatrix} e_3 & f_3 \\ g_3 & h_3 \end{bmatrix} = \begin{bmatrix} e_1 & f_1 \\ g_1 & h_1 \end{bmatrix} \cdot \begin{bmatrix} e_2 & f_2 \\ g_2 & h_2 \end{bmatrix}.$$

Proof. By leveraging the additivity of the DLCT and the GLCT, we observe that $\mathbf{O}_{\mathcal{D}}^{\mathbf{K}_3}=\mathbf{O}_{\mathcal{D}}^{\mathbf{K}_1}\cdot\mathbf{O}_{\mathcal{D}}^{\mathbf{K}_2}$ and $\mathbf{O}_{\mathcal{G}}^{\mathbf{M}_3}=\mathbf{O}_{\mathcal{G}}^{\mathbf{M}_1}\cdot\mathbf{O}_{\mathcal{G}}^{\mathbf{M}_2}$. Hence, we have

$$\begin{aligned}\mathbf{J}^{\mathbf{K}_3,\mathbf{M}_3} &= \mathbf{J}^{\mathbf{K}_1\mathbf{K}_2,\mathbf{M}_1\mathbf{M}_2} = \mathbf{O}_{\mathcal{D}}^{\mathbf{K}_1\mathbf{K}_2}\otimes\mathbf{O}_{\mathcal{G}}^{\mathbf{M}_1\mathbf{M}_2}\\ &= (\mathbf{O}_{\mathcal{D}}^{\mathbf{K}_1}\cdot\mathbf{O}_{\mathcal{D}}^{\mathbf{K}_2})\otimes(\mathbf{O}_{\mathcal{G}}^{\mathbf{M}_1}\cdot\mathbf{O}_{\mathcal{G}}^{\mathbf{M}_2})\\ &\overset{(a)}{=} (\mathbf{O}_{\mathcal{D}}^{\mathbf{K}_1}\otimes\mathbf{O}_{\mathcal{G}}^{\mathbf{M}_1})\cdot(\mathbf{O}_{\mathcal{D}}^{\mathbf{K}_2}\otimes\mathbf{O}_{\mathcal{G}}^{\mathbf{M}_2})\\ &= \mathbf{J}^{\mathbf{K}_1,\mathbf{M}_1}\cdot\mathbf{J}^{\mathbf{K}_2,\mathbf{M}_2},\end{aligned}$$

where equality (a) is derived from the properties of Kronecker product mixed products. □

Property 2. ***Commutativity:*** *For the LCT, commutativity is generally not satisfied. However, in the case of two parameter matrix pairs* $(\mathbf{K}_1,\mathbf{M}_1)$ *and* $(\mathbf{K}_2,\mathbf{M}_2)$*, if they fulfill the conditions* $\mathbf{K}_1\mathbf{K}_2=\mathbf{K}_2\mathbf{K}_1$ *and* $\mathbf{M}_1\mathbf{M}_2=\mathbf{M}_2\mathbf{M}_1$*, the JLCT exhibits commutative properties, such that*

$$\mathbf{J}^{\mathbf{K}_1,\mathbf{M}_1}\cdot\mathbf{J}^{\mathbf{K}_2,\mathbf{M}_2}=\mathbf{J}^{\mathbf{K}_2,\mathbf{M}_2}\cdot\mathbf{J}^{\mathbf{K}_1,\mathbf{M}_1} \tag{22}$$

and the JLCT is cross-commutative such that

$$\mathbf{J}^{\mathbf{K}_1,\mathbf{M}_1}\cdot\mathbf{J}^{\mathbf{K}_2,\mathbf{M}_2}=\mathbf{J}^{\mathbf{K}_1,\mathbf{M}_2}\cdot\mathbf{J}^{\mathbf{K}_2,\mathbf{M}_1}=\mathbf{J}^{\mathbf{K}_2,\mathbf{M}_1}\cdot\mathbf{J}^{\mathbf{K}_1,\mathbf{M}_2}. \tag{23}$$

Proof. This property follows the additivity of the JLCT

$$\mathbf{J}^{\mathbf{K}_1,\mathbf{M}_1}\cdot\mathbf{J}^{\mathbf{K}_2,\mathbf{M}_2}=\mathbf{J}^{\mathbf{K}_1\mathbf{K}_2,\mathbf{M}_1\mathbf{M}_2}=\mathbf{J}^{\mathbf{K}_2\mathbf{K}_1,\mathbf{M}_2\mathbf{M}_1}=\mathbf{J}^{\mathbf{K}_2,\mathbf{M}_2}\cdot\mathbf{J}^{\mathbf{K}_1,\mathbf{M}_1}$$

The cross-commutativity property also follows from the additivity of the JLCT

$$\begin{aligned}\mathbf{J}^{\mathbf{K}_1,\mathbf{M}_1}\cdot\mathbf{J}^{\mathbf{K}_2,\mathbf{M}_2} &= \mathbf{J}^{\mathbf{K}_1\mathbf{K}_2,\mathbf{M}_1\mathbf{M}_2}=\mathbf{J}^{\mathbf{K}_1\mathbf{K}_2,\mathbf{M}_2\mathbf{M}_1}\\ &= \mathbf{J}^{\mathbf{K}_1,\mathbf{M}_2}\cdot\mathbf{J}^{\mathbf{K}_2,\mathbf{M}_1}=\mathbf{J}^{\mathbf{K}_2\mathbf{K}_1,\mathbf{M}_1\mathbf{M}_2}\\ &= \mathbf{J}^{\mathbf{K}_2,\mathbf{M}_1}\cdot\mathbf{J}^{\mathbf{K}_1,\mathbf{M}_2}\end{aligned}$$

where the last equality comes from the commutativity under special conditions. □

Property 3. ***Invertibility:*** *The invertibility of the JLCT implies that the inverse JLCT (IJLCT) can be achieved through another JLCT characterized by parameter matrices equal to the inverse of the original transformation matrices. This property of invertibility is derived from the previously discussed additivity. For a parameter matrix pair* $(\mathbf{K},\mathbf{M})$*, it holds that*

$$\left(\mathbf{J}^{\mathbf{K},\mathbf{M}}\right)^{-1}\cdot\mathbf{J}^{\mathbf{K},\mathbf{M}}=\mathbf{J}^{\mathbf{K}^{-1},\mathbf{M}^{-1}}\cdot\mathbf{J}^{\mathbf{K},\mathbf{M}}=\mathbf{I}, \tag{24}$$

where $\mathbf{I}$ *denotes the identity matrix.*

Proof. Based on the invertibility of the DLCT and the GLCT

$$\begin{aligned}\left(\mathbf{J}^{\mathbf{K},\mathbf{M}}\right)^{-1} &= \left(\mathbf{O}_{\mathcal{D}}^{\mathbf{K}}\otimes\mathbf{O}_{\mathcal{G}}^{\mathbf{M}}\right)^{-1}=\left(\mathbf{O}_{\mathcal{D}}^{\mathbf{K}}\right)^{-1}\otimes\left(\mathbf{O}_{\mathcal{G}}^{\mathbf{M}}\right)^{-1}\\ &= \mathbf{O}_{\mathcal{D}}^{\mathbf{K}^{-1}}\otimes\mathbf{O}_{\mathcal{G}}^{\mathbf{M}^{-1}}=\mathbf{J}^{\mathbf{K}^{-1},\mathbf{M}^{-1}},\end{aligned} \tag{25}$$

demonstrating its invertibility through the identity matrix and leveraging the additive properties of JLCT indices. □

Remark 4. The inverse transform of the JLCT can be obtained through its invertibility, as follows

$$\mathbf{x}=\left(\mathbf{J}^{\mathbf{K},\mathbf{M}}\right)^{-1}\hat{\mathbf{x}}=\left(\mathbf{O}_{\mathcal{D}}^{\mathbf{K}^{-1}}\otimes\mathbf{O}_{\mathcal{G}}^{\mathbf{M}^{-1}}\right)\hat{\mathbf{x}}, \tag{26}$$

where the first equality on the right follows from the invertibility of the Kronecker product, that is, $\left(\mathbf{O}_{\mathcal{D}}^{\mathbf{K}}\otimes\mathbf{O}_{\mathcal{G}}^{\mathbf{M}}\right)^{-1}=\mathbf{O}_{\mathcal{D}}^{\mathbf{K}^{-1}}\otimes\mathbf{O}_{\mathcal{G}}^{\mathbf{M}^{-1}}$.

Property 4. ***Separability:*** *The JLCT is separable on the GLCT and the DLCT. That is, for a given parameter matrix pair* $(\mathbf{K},\mathbf{M})$*, we have*

$$\mathbf{J}^{\mathbf{K},\mathbf{M}}=\mathbf{J}^{\mathbf{K},\mathbf{I}}\cdot\mathbf{J}^{\mathbf{I},\mathbf{M}}=\mathbf{J}^{\mathbf{I},\mathbf{M}}\cdot\mathbf{J}^{\mathbf{K},\mathbf{I}}. \tag{27}$$

Proof. The first equality follows from additivity, while the second follows from special commutativity. By incorporating Eqs. (19) and (20), we can also obtain $\mathbf{J}^{\mathbf{K},\mathbf{I}}=\mathbf{O}_{\mathcal{D}}^{\mathbf{K}}$ and $\mathbf{J}^{\mathbf{I},\mathbf{M}}=\mathbf{O}_{\mathcal{G}}^{\mathbf{M}}$. In particular, using reduction from the identity properties of the GLCT and DLCT and the definition of the JLCT, we obtain Eq. (27) as

$$\begin{aligned}\mathbf{J}^{\mathbf{K},\mathbf{M}} &= \mathbf{O}_{\mathcal{D}}^{\mathbf{K}}\otimes\mathbf{O}_{\mathcal{G}}^{\mathbf{M}}\\ &= (\mathbf{O}_{\mathcal{D}}^{\mathbf{K}}\otimes\mathbf{I}_T)\otimes\left(\mathbf{O}_{\mathcal{G}}^{\mathbf{M}}\otimes\mathbf{I}_N\right)\\ &= \mathbf{O}_{\mathcal{D}}^{\mathbf{K}}\otimes\left(\mathbf{I}_T\otimes\mathbf{O}_{\mathcal{G}}^{\mathbf{M}}\otimes\mathbf{I}_N\right)\\ &= (\mathbf{I}_T\otimes\mathbf{O}_{\mathcal{D}}^{\mathbf{K}}\otimes\mathbf{I}_N)\otimes\mathbf{O}_{\mathcal{G}}^{\mathbf{M}}.\end{aligned}$$

This clearly demonstrates the separability of the JLCT. □

Property 5. ***Reducibility:*** *The JFRFT becomes a special case of the JLCT, when*

$$\mathbf{K}=\begin{bmatrix}\cos\alpha & \sin\alpha\\ -\sin\alpha & \cos\alpha\end{bmatrix}, \text{and } \mathbf{M}=\begin{bmatrix}\cos\beta & \sin\beta\\ -\sin\beta & \cos\beta\end{bmatrix}.$$

The JFT becomes a special case of the JLCT, when $\mathbf{K}=\mathbf{M}=\begin{bmatrix}0 & 1\\ -1 & 0\end{bmatrix}$ *or* $\alpha=\beta=1$.

Proof. According to the reducibility of DLCT and GLCT, we can get

$$\begin{aligned}\text{JFRFT}: \mathbf{J}^{\mathbf{K},\mathbf{M}} &= \mathbf{O}_{\mathcal{D}}^{\mathbf{K}}\otimes\mathbf{O}_{\mathcal{G}}^{\mathbf{M}}=\mathbf{F}_{\mathcal{D}}^{\alpha}\otimes\mathbf{F}_{\mathcal{G}}^{\beta}=\mathbf{J}^{\alpha,\beta},\\ \text{JFT}: \mathbf{J}^{\mathbf{K},\mathbf{M}} &= \mathbf{O}_{\mathcal{D}}^{\mathbf{K}}\otimes\mathbf{O}_{\mathcal{G}}^{\mathbf{M}}=\mathbf{F}_{\mathcal{D}}^{1}\otimes\mathbf{F}_{\mathcal{G}}^{1}=\mathbf{J}.\end{aligned} \tag{28}$$

Here, $\mathbf{O}_{\mathcal{D}}^{\mathbf{K}}=\mathbf{D}^0\mathbf{E}_1\mathbf{D}^{\alpha}\mathbf{E}^{-1}=\mathbf{F}_{\mathcal{D}}^{\alpha}$ and $\mathbf{O}_{\mathcal{G}}^{\mathbf{M}}=\mathbf{\Lambda}^0\mathbf{Q}_1\mathbf{\Lambda}^{\beta}\mathbf{Q}^{-1}=\mathbf{F}_{\mathcal{G}}^{\beta}$. □

Property 6. ***Identity:*** *When* $\mathbf{K}=\mathbf{M}=\mathbf{I}$*,* $\mathbf{J}^{\mathbf{I},\mathbf{I}}=\mathbf{O}_{\mathcal{D}}^{\mathbf{I}}\otimes\mathbf{O}_{\mathcal{G}}^{\mathbf{I}}=\mathbf{I}$*, which is the reduction to the identity matrix.*

Proof. This proof follows from definition of the JLCT and the simplification of the identity properties of the DLCT and the GLCT, i.e. $\mathbf{O}_{\mathcal{D}}^{\mathbf{I}}=\mathbf{D}^0\mathbf{E}_1\mathbf{D}^0\mathbf{E}^{-1}=\mathbf{I}$ and $\mathbf{O}_{\mathcal{G}}^{\mathbf{I}}=\mathbf{\Lambda}^0\mathbf{Q}_1\mathbf{\Lambda}^0\mathbf{Q}^{-1}=\mathbf{I}$. □

Property 7. ***Unitary:*** *The JLCT is a unitary transform, or equivalently,* $\mathbf{J}^{\mathbf{K},\mathbf{M}}$ *is a unitary matrix.*

Proof. Leveraging the properties of the Kronecker product, if $\mathbf{O}_{\mathcal{D}}^{\mathbf{K}}$ and $\mathbf{O}_{\mathcal{G}}^{\mathbf{M}}$ are unitary matrices, then $\mathbf{J}^{\mathbf{K},\mathbf{M}}$ is also a unitary matrix. The properties of the DLCT [40] demonstrate that $\mathbf{O}_{\mathcal{D}}^{\mathbf{K}}$ is unitary for any parameter matrix $\mathbf{K}$. Similarly, the properties of the GLCT [46] indicate that $\mathbf{O}_{\mathcal{G}}^{\mathbf{M}}$ is unitary. Consequently, the JLCT constitutes a unitary transform as well. □

Remark 5. When $\mathbf{A}=\mathbf{C}$, the underlying graph is a cycle graph [50], which is consistent with the description in Appendix A. Within the framework of JLCT, the matrices of GLCT and DLCT coincide. Thus, in joint time-vertex signal processing, if $N=T$, the adjacency matrix $\mathbf{A}\in\mathbb{C}^{N\times N}$ is equivalent to the matrix $\mathbf{C}\in\mathbb{C}^{T\times T}$, where

$$\mathbf{C}=\begin{bmatrix} & & & 1\\ 1 & & & \\ & \ddots & & \\ & & 1 & \end{bmatrix}.$$

Then, we have the following corollaries.

Corollary 1. *If the underlying graph is a directed cyclic graph ($\sigma = \varsigma = 1$), then $\mathbf{J}^{\mathbf{K},\mathbf{M}}$ reduces to the two-dimensional DLCT with parameter matrices* $\mathbf{K}$ *and* $\mathbf{M}$.

Proof. For $\mathbf{J}^{\mathbf{K},\mathbf{M}} = \mathbf{O}_{\mathcal{D}}^{\mathbf{K}} \otimes \mathbf{O}_{\mathcal{G}}^{\mathbf{M}}$, given that the underlying graph is cyclic, as indicated in **Remark 5**, it can be deduced that $\mathbf{O}_{\mathcal{D}}^{\mathbf{K}}$ is equivalent to $\mathbf{O}_{\mathcal{G}}^{\mathbf{M}}$. The operator matrix $\mathbf{O}_{\mathcal{G}}^{\mathbf{M}}$ can be rewritten as

$$\mathbf{O}_{\mathcal{G}}^{\mathbf{M}} = \mathbf{D}^{\epsilon}\mathbf{E}_{\varsigma}\mathbf{D}^{\beta}\mathbf{E}^{-1} = \mathbf{O}_{\mathcal{D}}^{\mathbf{M}}, \tag{29}$$

where the definitions of $\mathbf{D}^{\epsilon}$, $\mathbf{E}_{\varsigma}$, $\mathbf{D}^{\beta}$, and $\mathbf{E}^{-1}$ are provided in Eq. (13). Therefore, we derive that $\mathbf{J}^{\mathbf{K},\mathbf{M}} = \mathbf{O}_{\mathcal{D}}^{\mathbf{K}} \otimes \mathbf{O}_{\mathcal{D}}^{\mathbf{M}}$ is equivalent to the two-dimensional DLCT. □

Corollary 2. *If we conceptualize the time domain as a graph structure as well, thereby perceiving DLCT as a specialized form of GLCT, then $\mathbf{J}^{\mathbf{K},\mathbf{M}}$ represents a two-dimensional GLCT characterized by the parameter matrices* $\mathbf{K}$ *and* $\mathbf{M}$.

Proof. Given that the JLCT adheres to the fundamental properties of the GLCT, and considering the DLCT as a particular instance of the GLCT within the framework of a directed cyclic graph structure, parallel to the proof presented in **Corollary 1**, we have that $\mathbf{O}_{\mathcal{D}}^{\mathbf{K}} = \mathbf{\Lambda}^{\xi}\mathbf{Q}_{\sigma}\mathbf{\Lambda}^{\alpha}\mathbf{Q}^{-1} = \mathbf{O}_{\mathcal{G}}^{\mathbf{K}}$, which leads to the conclusion that $\mathbf{J}^{\mathbf{K},\mathbf{M}} = \mathbf{O}_{\mathcal{G}}^{\mathbf{K}} \otimes \mathbf{O}_{\mathcal{G}}^{\mathbf{M}}$. This perspective offers a method for extending the multidimensional DLCT into the domain of GSP. □

As a two-dimensional transform, the JLCT shares several properties with the two-dimensional DLCT, including additivity, invertibility, reducibility, identity reduction, and unitary. These properties are naturally extended to the two-dimensional GLCT as well.

The aforementioned properties and corollaries demonstrate that both JFT and JLCT possess invertibility, separability, identity, and unitarity. However, due to the lack of flexibility in JFT compared to JLCT, the latter also exhibits additivity, reducibility, and commutativity. In relation to the underlying graphs, both transforms can be converted into two-dimensional transforms under certain special forms. These differences highlight that JLCT extends the concept of JFT by incorporating properties dependent on parameter matrices, making it more flexible and suitable for a broader range of transformations, including those involving graph structures.

4. Numerical experiments and applications

The JLCT framework for time-vertex signal processing demonstrates significant applications on dynamic mesh datasets. We employed the JLCT in numerical examples for two distinct tasks: denoising and clustering [22,35], comparing the energy compression of different JLCT parameter settings within the dataset. Numerical experiments indicate that the JLCT enhances regression and learning tasks, refining and augmenting the performance of JFT and JFRFT. The experiments were conducted using tools from GSPBox [51].

4.1. Denoise of dog walking dynamic mesh

In this section, we proceed with denoising on dynamic meshes to evaluate the practicality of the JLCT for regression problems. Assuming the dynamic mesh model is smooth, for the denoising methods of the JFT [22] and the JFRFT [35], they employ the joint Tikhonov regularization method. Since this method is defined using the Laplacian operator, it is not applicable to JLCT. Therefore, we use a new filter-based method for denoising. This approach extends time-vertex signal processing and achieves effective denoising results. Here, we introduce a filter based on perfect localization [12].

In the time domain, we utilize the GLCT on cycle graphs, specifically the DLCT, for filter construction. Assume the signal $\mathbf{X} \in \mathbb{C}^{N\times T}$ is to be preprocessed using the spectral domain filter $\mathbf{\Sigma}$. Initially, the DLCT is applied to $\mathbf{X}$, yielding $\hat{\mathbf{X}} = \mathbf{X}(\mathbf{O}_{\mathcal{D}}^{\mathbf{K}})^{\mathrm{H}}$. Subsequently, $\hat{\mathbf{X}}$ is filtered as follows

$$\hat{\mathbf{Y}} = \hat{\mathbf{X}}\mathbf{\Sigma} = \hat{\mathbf{X}}\mathrm{diag}\,(1, ..., 1)\,,$$

where $\mathbf{\Sigma}$ is a diagonal matrix defined such that $\mathbf{\Sigma}_{ii} = 1$ if $i \in \mathcal{F}$, and $\mathbf{\Sigma}_{ii} = 0$ if $i \notin \mathcal{F}$. The index set $\mathcal{F}$ represents a subset of the spectrum, with the condition $|\mathcal{F}| \le T$. Furthermore, the inverse DLCT is applied to $\hat{\mathbf{Y}}$, resulting in $\mathbf{Y} = \hat{\mathbf{Y}}\mathbf{O}_{\mathcal{D}}^{\mathbf{K}}$. Combining these steps, we obtain

$$\mathbf{Y} = \mathbf{X}(\mathbf{O}_{\mathcal{D}}^{\mathbf{K}})^{\mathrm{H}}\mathbf{\Sigma}\mathbf{O}_{\mathcal{D}}^{\mathbf{K}}, \tag{30}$$

where the filter can be defined as $\mathbf{B}^{\mathbf{K}} = (\mathbf{O}_{\mathcal{D}}^{\mathbf{K}})^{\mathrm{H}}\mathbf{\Sigma}\mathbf{O}_{\mathcal{D}}^{\mathbf{K}}$.

This method allows for filtering in the frequency domain to perform various information processing tasks, achieving denoised signals in the time domain via DLCT, denoted as $\mathbf{Y}$. Thus, akin to the time domain, we utilize the GLCT to define the filter in the vertex domain. The perfectly localized signal is given by

$$\mathbf{Y} = (\mathbf{O}_{\mathcal{G}}^{\mathbf{M}})^{\mathrm{H}}\mathbf{\Pi}\mathbf{O}_{\mathcal{G}}^{\mathbf{M}}\mathbf{X}, \tag{31}$$

where the filter in the vertex domain is defined as $\mathbf{B}^{\mathbf{M}} = (\mathbf{O}_{\mathcal{G}}^{\mathbf{M}})^{\mathrm{H}}\mathbf{\Pi}\mathbf{O}_{\mathcal{G}}^{\mathbf{M}}$. Here, $\mathbf{\Pi}$ is defined analogously to $\mathbf{\Sigma}$, representing the frequency response of the graph filter, $\mathbf{\Pi}_{jj} = 1$ if $j \in \mathcal{H}$, and $\mathbf{\Pi}_{jj} = 0$ if $j \notin \mathcal{H}$, with $|\mathcal{H}| \le N$.

After obtaining filters in both the time and vertex domains, we can achieve a perfectly localized signal under the joint time-vertex signal processing framework

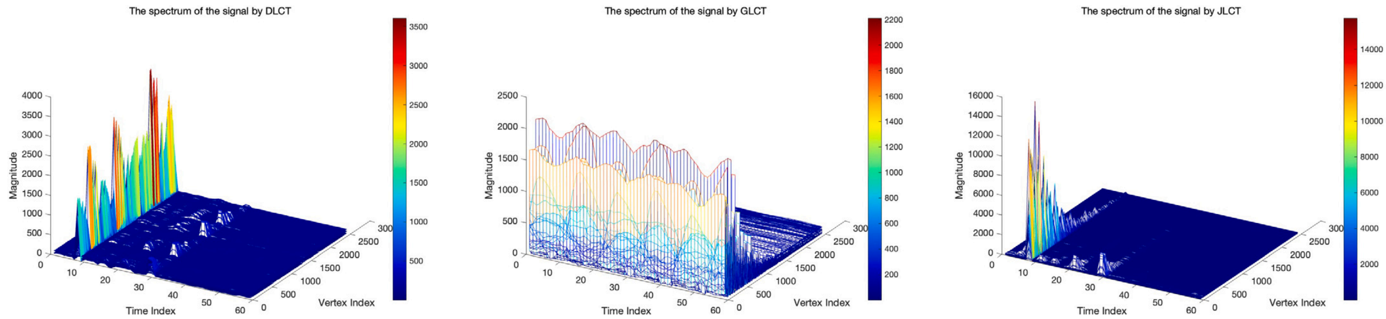
$$\mathbf{Y} = \mathbf{B}^{\mathbf{M}}\mathbf{X}\mathbf{B}^{\mathbf{K}} = (\mathbf{O}_{\mathcal{G}}^{\mathbf{M}})^{\mathrm{H}}\mathbf{\Pi}\mathbf{O}_{\mathcal{G}}^{\mathbf{M}}\mathbf{X}(\mathbf{O}_{\mathcal{D}}^{\mathbf{K}})^{\mathrm{H}}\mathbf{\Sigma}\mathbf{O}_{\mathcal{D}}^{\mathbf{K}}. \tag{32}$$

To perform joint denoising, we utilize filters in both the time and vertex domains. Initially, we preprocess the dog walking dynamic mesh dataset to study its performance. We consider the vertices of a 3D time-varying mesh with dimensions $2502 \times 59 \times 3$ ($N \times T \times$ dimensions). We construct a graph signal based on the distances between coordinate-based temporal averages, as illustrated in Fig. 4(a). Applying Eqs (5), (13), and (16), we also present the spectral representations in the time domain and the vertex domain, as well as the joint time-vertex domain under different parameter matrices $\mathbf{K} = [0, 2; -1/2, 2]$ and $\mathbf{M} = [0, 1/2; -2, -1/2]$, as shown in Fig. 3.

We augment the coordinates of the dog walking dynamic mesh dataset with noise of the same size as the original signal, drawn randomly from a standard normal distribution. To maintain the relative strength of the noise, the norm of the noise matrix is scaled to 20% of the original signal, followed by the application of filtering to each coordinate dimension using Eqs. (30)-(32). We choose $|\mathcal{F}| = 30, |\mathcal{H}| = 1503$. The denoised graph signals are illustrated in Fig. 4. (b) presents the graph signal denoised using a DLCT-based filter characterized by the parameter matrix $\mathbf{K}$ in the time domain. (c) depicts the graph signal denoised through a GLCT-based filter determined by the parameter matrix $\mathbf{M}$ in the vertex domain. (d) shows the graph signal after denoising with a combined application of the filters from (b) and (c).

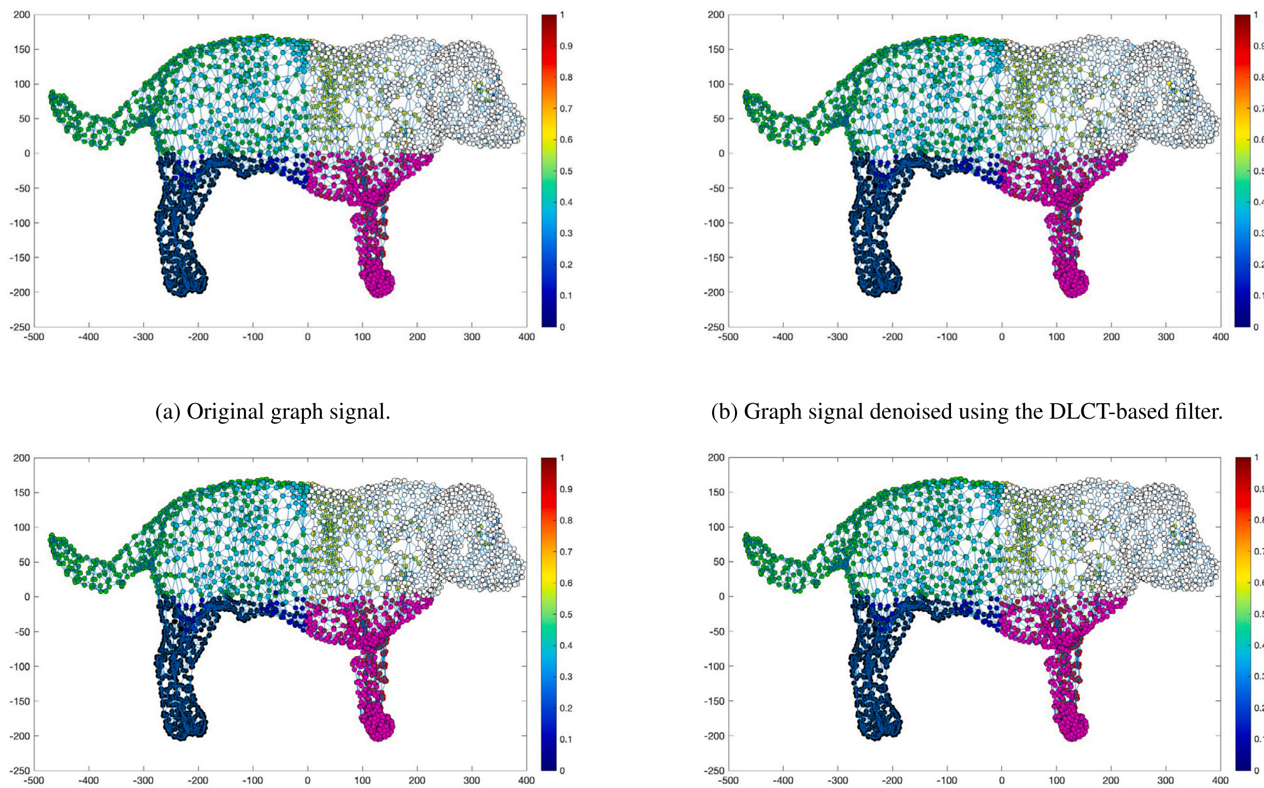
Subsequently, we compare the denoising effects of filters based on DLCT, GLCT, and JLCT under different parameter matrices $\mathbf{K}$ and $\mathbf{M}$, utilizing different baselines. We employ the normalized mean square error (NMSE) to assess discrepancies between the original and the denoised signals, where the NMSE between the original and noisy signals is 0.04.

Initially, we compare different $\mathbf{K}$ matrices in the time domain, setting the matrix $\mathbf{M}$ to the identity matrix $[1, 0; 0, 1]$. Given that the parameter ξ solely alters the real and imaginary parts without affecting the magnitude, we fix $\xi = 1$. As per [37], the performance improves as α approaches 1; hence, we vary α from 0.9 to 1. With parameters



(a) DLCT in time domain. (b) GLCT in vertex domain. (c) JLCT in joint time-vertex domain.

Fig. 3. Spectrum of the dog walking dynamic mesh dataset signal.



(a) Original graph signal. (b) Graph signal denoised using the DLCT-based filter.

(c) Graph signal denoised using the GLCT-based filter. (d) Graph signal denoised using the JLCT-based filter.

Fig. 4. Representation of the original and denoised graph signals from the dog walking dynamic mesh dataset.

$\sigma = m \times 2^n, n \in \mathbb{R}$, setting $m = 1, 3, 5, 7, N$, as illustrated in Fig. 5(a), we observe that the smallest NMSE of 0.0217 occurs when $\alpha = 1$ and $\sigma = 2^n$.

Subsequently, we evaluate different $\mathbf{M}$ matrices in the vertex domain, again setting $\mathbf{K}$ to the identity matrix. Similar to the time domain, we fix $\epsilon = -1$ and let β vary from 0.9 to 1 with $\varsigma = m \times 2^n$. As shown in Fig. 5(b), the minimal NMSE of 0.0282 is achieved when $\beta = 1$ and $\varsigma = 2^n$.

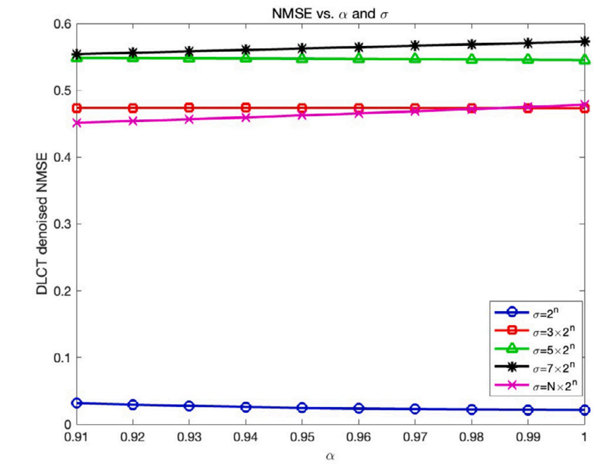
Finally, in the joint time-vertex domain, we compare different $\mathbf{K}$ and $\mathbf{M}$ matrices while keeping $\xi = 1$ and $\epsilon = -1$, and varying α and β from 0.9 to 1. Fig. 6(a) displays cases where $\sigma = \varsigma = 2^n$, (b) shows $\sigma = 2^n$, $\varsigma = N \times 2^n$, (c) represents $\sigma = T \times 2^n$, $\varsigma = 2^n$, and (d) depicts $\sigma = T \times 2^n$, $\varsigma = N \times 2^n$. Notably, when both $\mathbf{K}$ and $\mathbf{M}$ matrices are optimized, the smallest NMSE of 0.0178 is attained using JLCT, outperforming the results from using DLCT and GLCT alone. We have summarized the parameters from Figs. 5 and 6 in Table 1 to avoid any confusion, in which '–' indicates unneeded.

Table 1
Variation of parameters in dog walking dynamic mesh denoising, where $m = 1, 3, 5, 7, N, i = 1, T, j = 1, N$.

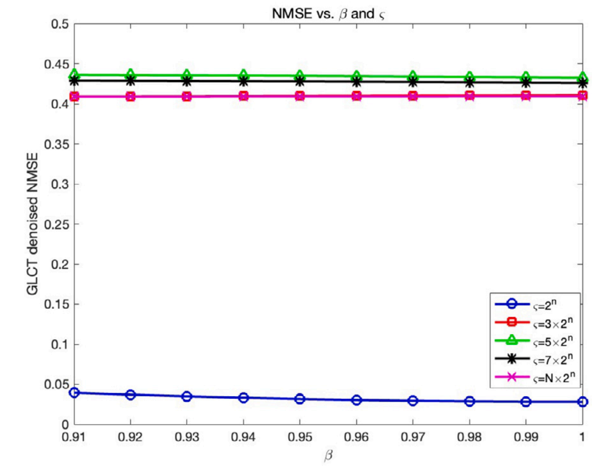
Transforms	ξ	σ	α	ϵ	ς	β
DLCT	1	$m \times 2^n$	$0.9 \to 1$	–	–	–
GLCT	–	–	–	-1	$m \times 2^n$	$0.9 \to 1$
JLCT	1	$i \times 2^n$	$0.9 \to 1$	-1	$j \times 2^n$	$0.9 \to 1$

4.2. Clustering of dancing man dynamic mesh

In this experiment, we conducted a preprocessing of the motion dataset from the dancing man dynamic mesh to evaluate feature ex-

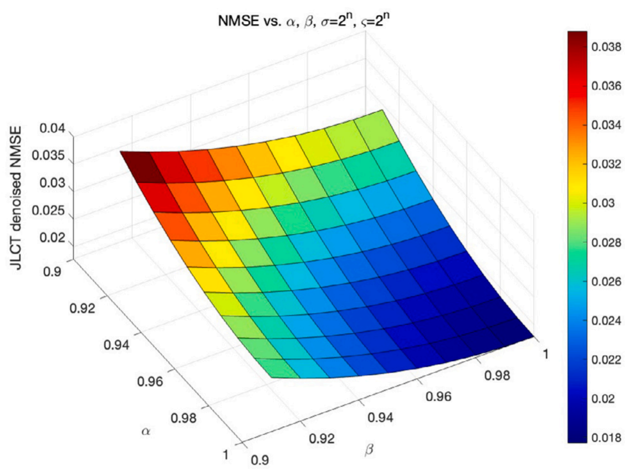


(a) NMSE vs. varying DLCT parameters in the time domain for denoising. Different colored curves represent various values of parameter σ, illustrating the change in error as α varies from 0.9 to 1.

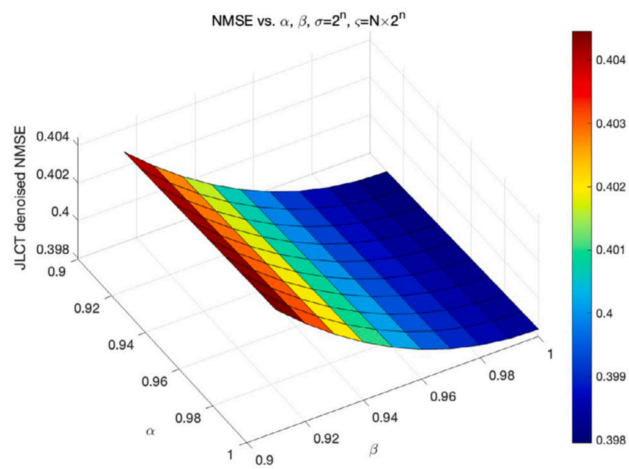


(b) NMSE vs. varying GLCT parameters in the vertex domain for denoising. Different colored curves represent various values of parameter ς, illustrating the change in error as β varies from 0.9 to 1.

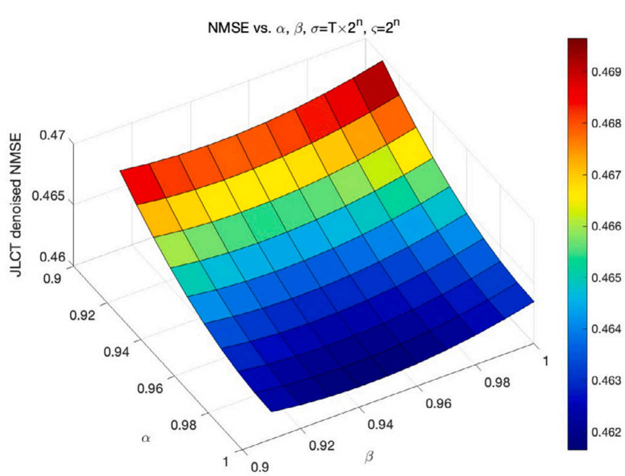
Fig. 5. NMSE vs. varying parameters in the time and vertex domains for denoising.



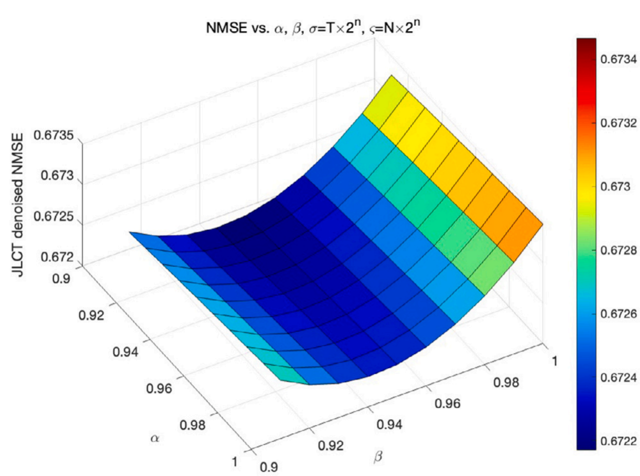
(a) NMSE vs. different values of α and β (ranging from 0 to 1), under $\sigma = \varsigma = 2^n$ conditions.



(b) NMSE vs. different values of α and β (ranging from 0 to 1), under $\sigma = 2^n, \varsigma = N \times 2^n$ conditions.

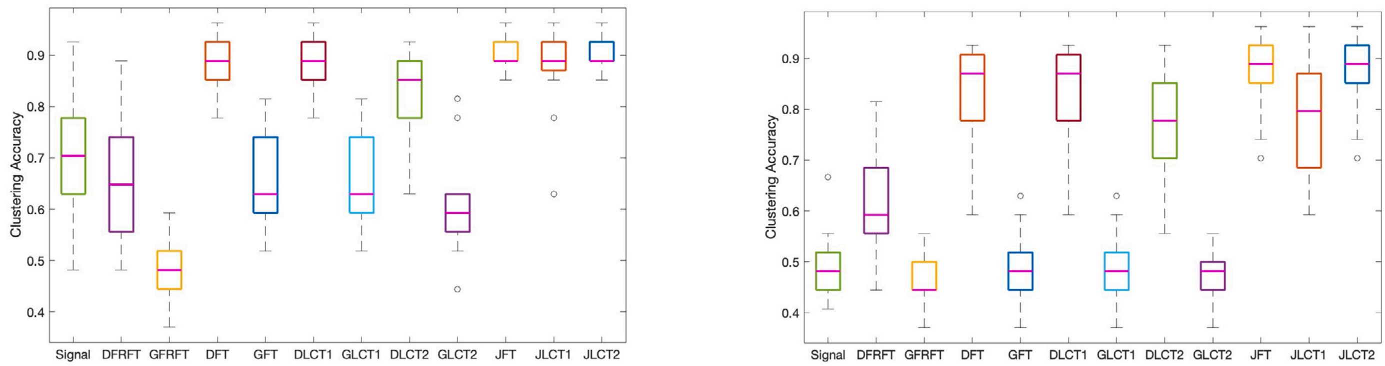


(c) NMSE vs. different values of α and β (ranging from 0 to 1), under $\sigma = T \times 2^n, \varsigma = 2^n$ conditions.



(d) NMSE vs. different values of α and β (ranging from 0 to 1), under $\sigma = T \times 2^n, \varsigma = N \times 2^n$ conditions.

Fig. 6. NMSE vs. varying JLCT parameters in the joint time-vertex domain for denoising.



(a) Sparse Gaussian noise is SNR -10 dB.

(b) Sparse Gaussian noise is SNR -20 dB.

Fig. 7. Comparison of clustering accuracy of the JLCT using different parameter matrices. DFRFT with $\alpha = 0.5$, GFRFT with $\beta = 0.5$, and the remaining LCT parameters are summarized in Table 2.

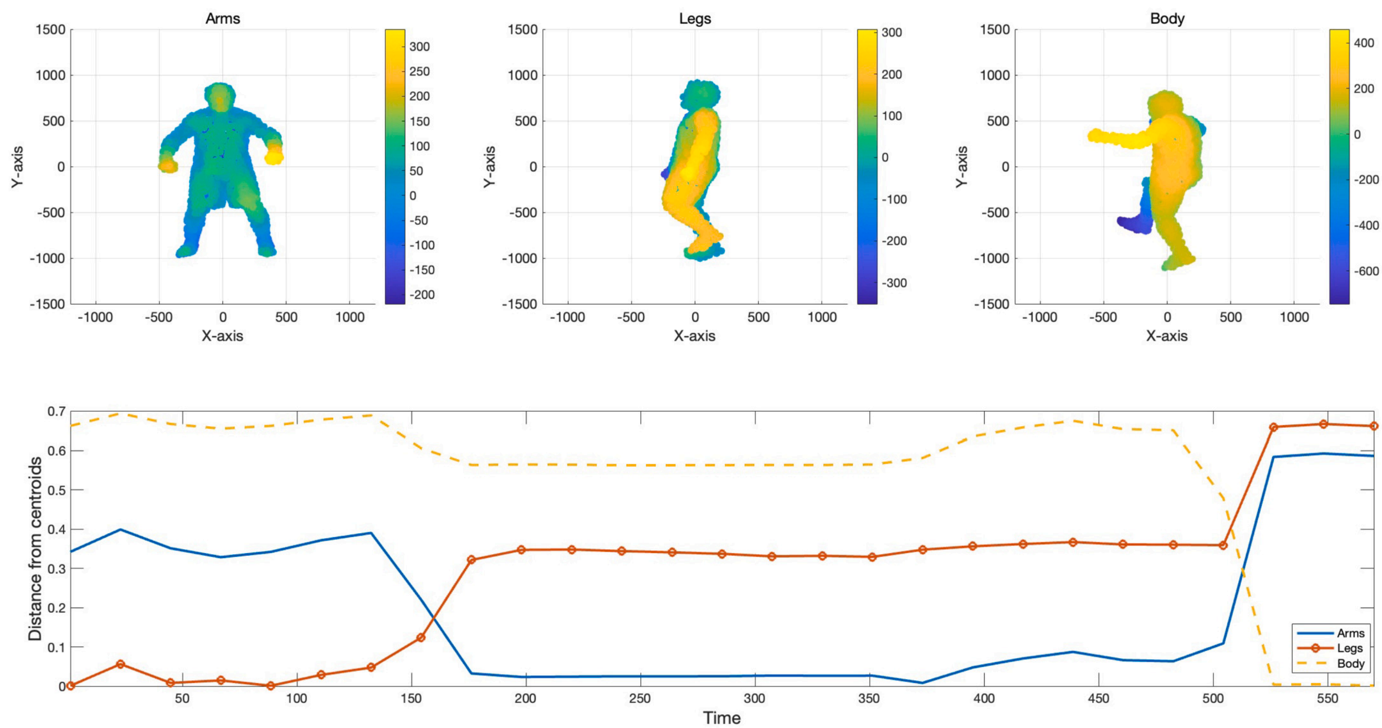
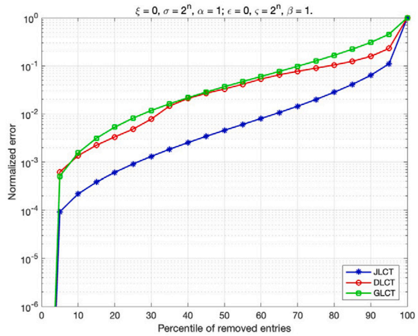


Fig. 8. Clustering of the dancing man dynamic mesh.

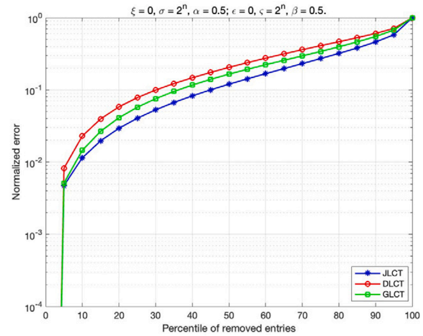
traction capabilities within the context of joint clustering, here using UNLocBox [52] and LTFAT [53]. We considered a 3D time-varying mesh of vertices with dimensions $1502 \times 573 \times 3$ ($N \times T \times$ dimensions). Our approach extends the experimental procedure demonstrated in [22] to showcase the performance and practicality of the proposed JLCT framework for clustering experiments. We introduced sparse noise with a density of 0.1, meaning approximately 10% of mesh points were subjected to corruption, with entries following a normal distribution. The noise was Gaussian, with SNR of -10 dB and -20 dB. A rectangular window size of 50, with a 60% overlap rate, and a temporal sequence of 27 were employed. A nearest-neighbor graph was utilized to capture the geometric structure of the data, ultimately facilitating the construction of graph signals.

We computed the JLCT for each coordinate dimension of the sequences obtained, concatenated the resulting joint time-vertex signals, and subsequently applied the k-means clustering algorithm to the representations for classification. The experiments were conducted 20 times to ensure robustness. Figs. 7(a) and (b) present the average accuracy results of the JLCT with different parameter matrices under -10 dB and -20 dB SNR, respectively. Our results demonstrate that the optimal parameter settings for DLCT, with $\xi = 1$, $\sigma = 1/2$, and $\alpha = 1$, and for GLCT, with $\epsilon = 0$, $\varsigma = 2$, and $\beta = 1$, enable the JLCT to achieve a maximum clustering accuracy of 0.9630. This underscores the efficacy of comparing clustering accuracies under various JLCT configurations and includes special baselines such as DFT, GFT, DFRFT, GFRFT, and JFT. The corresponding parameters are summarized in Table 2, in which '–' indicates unneeded.

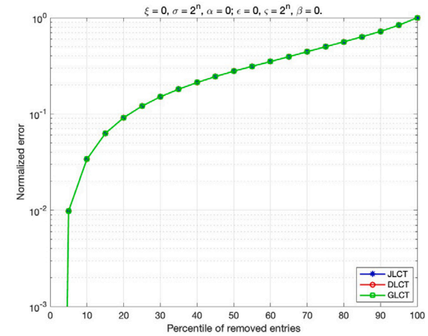
We utilized the optimal JLCT parameters to extract time-correlated features for classifying dance movements, as depicted in Fig. 8. These movements include arm, leg, and body motions, represented in two-dimensional plots derived from a subset of the dancer mesh dataset. The graph below plot illustrates the distances between points represented by the optimal JLCT and their respective clustering centroids. It was observed that each frame corresponds to distinct stages of the dance



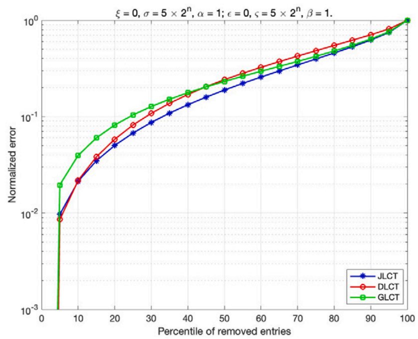
(a) $\xi = \epsilon = 0, \sigma = \varsigma = 2^n, \alpha = \beta = 1.$



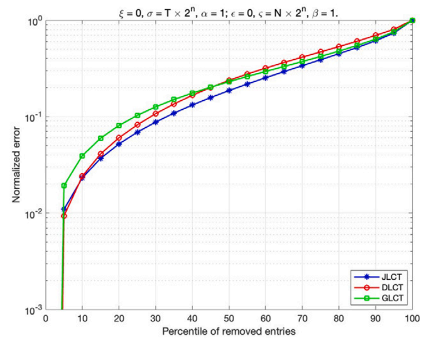
(b) $\xi = \epsilon = 0, \sigma = \varsigma = 2^n, \alpha = \beta = 0.5.$



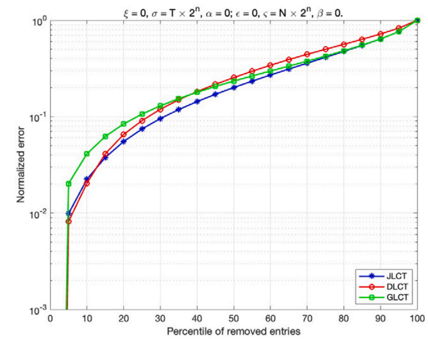
(c) $\xi = \epsilon = 0, \sigma = \varsigma = 2^n, \alpha = \beta = 0.$



(d) $\xi = \epsilon = 0, \sigma = \varsigma = 5 \times 2^n, \alpha = \beta = 1.$



(e) $\xi = \epsilon = 0, \sigma = T \times 2^n, \varsigma = N \times 2^n, \alpha = \beta = 1.$



(f) $\xi = \epsilon = 0, \sigma = T \times 2^n, \varsigma = N \times 2^n, \alpha = \beta = 0.$

Fig. 9. Energy compactness of the DLCT, the GLCT and the JLCT under different parameters in the dog walking dynamic mesh dataset.

Table 2
Variation of parameters in the dancing man dynamic mesh clustering.

Transforms	ξ	σ	α	ϵ	ς	β
DLCT1	0	4	1	–	–	–
GLCT1	–	–	–	0	2	1
DLCT2	1	1	1	–	–	–
GLCT2	–	–	–	-1	1	1
JLCT	1	1	1	-1	1	1
JLCT	1	1/2	1	0	2	1

sequence, with the JLCT using optimal parameters producing results that are comparable to or surpass those reported in [22].

4.3. Compactness comparison

To assess the performance of the JLCT in handling graph signals, we explore its capability to compactly encode the evolution of signals related to time-varying graphs. Our initial investigation involves analyzing the energy compaction properties of the JLCT in two datasets: the dog walking dynamic mesh as discussed in Section 4.1 and the dancing man dynamic mesh presented in Section 4.2.

To measure energy compaction, we compute the DLCT, the GLCT, and the JLCT for various parameter matrices within each dataset. We then zero out transform values below a percentile threshold p and calculate the compression error of the reconstructed signal (via the respective inverse transform) as follows

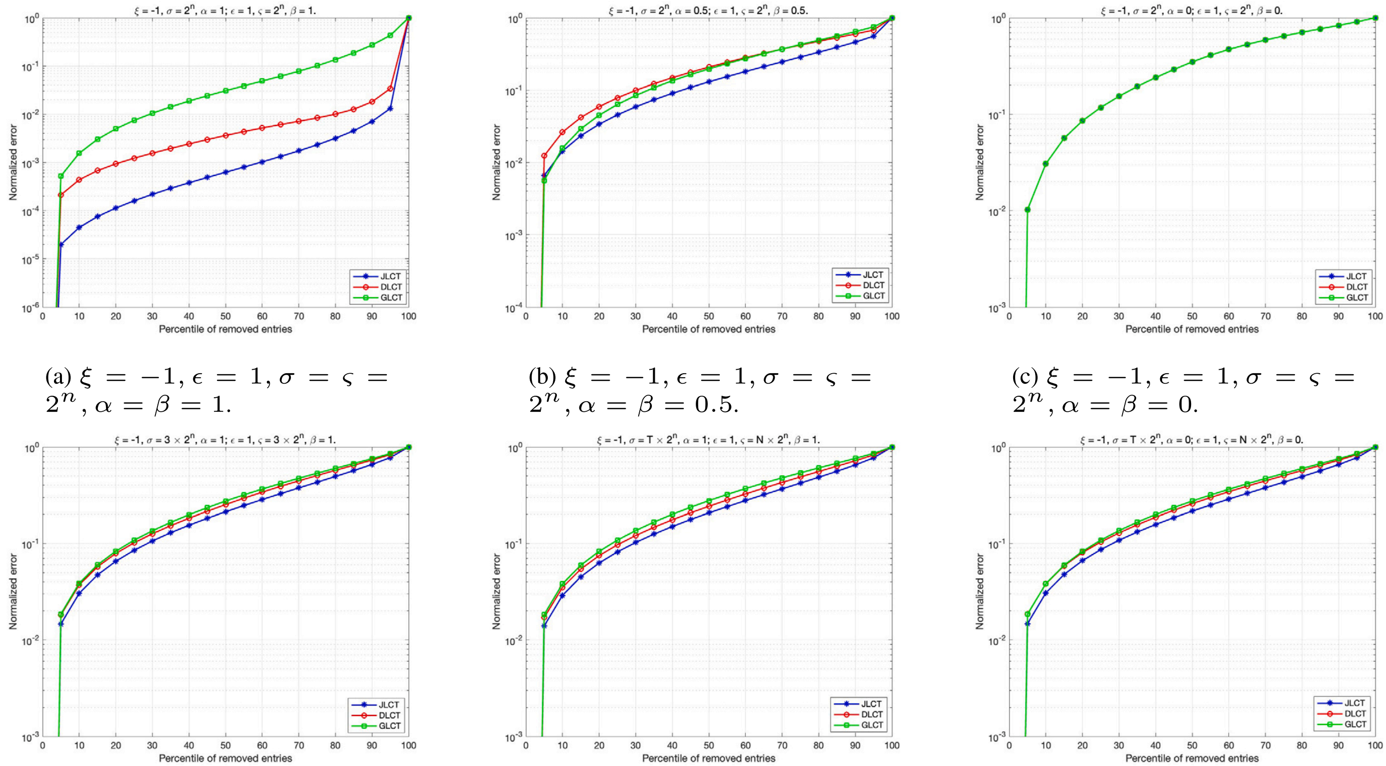
$$\text{error} = \frac{\|\mathbf{X} - \mathbf{X}_p\|_F}{\|\mathbf{X}\|_F}, \tag{33}$$

where $\mathbf{X}$ denotes the original signal, $\mathbf{X}_p$ represents the compressed signal, and F represents the Frobenius norm.

Fig. 9 illustrates the energy compaction characteristics of the DLCT, GLCT, and JLCT under varying parameters within the dog walking dynamic mesh dataset. Panel (a) represents a specific instance of the JLCT, including the JFT, while panels (b) and (c) correspond to instances of the JFRFT with orders $\alpha = \beta = 0.5$ and 0, respectively. Panels (d) and (e) depict energy compaction under different values of σ and ς with $\alpha = \beta = 1$, whereas panel (f) shows the effect of varying σ and ς when $\alpha = \beta = 0$. From panels (a) to (c), it is evident that reducing α and β tends to align the performance of the DLCT, GLCT, and JLCT. Panels (d) and (e) demonstrate that varying σ and ς can bring these transforms closer, with a more pronounced impact on the GLCT. Despite the convergence of the curves in panel (c) when $\alpha = \beta = 0$, varying σ and ς as shown in panel (f) still introduces noticeable differences. Fig. 10 compares the energy compaction characteristics of the DLCT, GLCT, and JLCT in the dancing man dynamic mesh dataset, with findings analogous to those observed in Fig. 9.

5. Conclusion

This paper introduces the joint time-vertex linear canonical transform (JLCT), a generalization of both the joint time-vertex Fourier transform (JFT) and the joint time-vertex fractional Fourier transform (JFRFT), grounded in the analysis of time-varying graph signals and their applications. Additionally, we redefine the discrete linear canonical transform for cyclic graphs and explore some properties of the JLCT, providing rigorous proofs for these characteristics. By extending Fourier analysis from the time and vertex domains to the domain of the linear canonical transform (LCT), we offer an enhanced set of the LCT analytical tools for joint time-vertex processing. The applications of the JLCT in dynamic mesh denoising, clustering, and energy compactness demonstrate that JLCT not only enhances regression and learning tasks but also refines and improves the performance of the JFT and the JFRFT.



(a) $\xi = -1, \epsilon = 1, \sigma = \varsigma = 2^n, \alpha = \beta = 1$.

(b) $\xi = -1, \epsilon = 1, \sigma = \varsigma = 2^n, \alpha = \beta = 0.5$.

(c) $\xi = -1, \epsilon = 1, \sigma = \varsigma = 2^n, \alpha = \beta = 0$.

(d) $\xi = -1, \epsilon = 1, \sigma = \varsigma = 3 \times 2^n, \alpha = \beta = 1$.

(e) $\xi = -1, \epsilon = 1, \sigma = T \times 2^n, \varsigma = N \times 2^n, \alpha = \beta = 1$.

(f) $\xi = -1, \epsilon = 1, \sigma = T \times 2^n, \varsigma = N \times 2^n, \alpha = \beta = 0$.

Fig. 10. Energy compactness of the DLCT, the GLCT and the JLCT under different parameters in the dancing man dynamic mesh dataset.

CRediT authorship contribution statement

Yu Zhang: Conceptualization, Data curation, Methodology, Software, Visualization, Writing – original draft. **Bing-Zhao Li:** Conceptualization, Funding acquisition, Project administration, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. DLCT corresponding to cycle graph

Given that the GFRFT can degenerate into the DFRFT in cyclic graphs, we redefine the DLCT starting from the discrete counterparts of the Hermite-Gaussian functions set [36] used in the DFRFT.

Hermite-Gaussian functions are well-known as eigenfunctions of the FT operator. We initiate from the defining Hermite-Gaussian differential equation

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}\Phi_p(t) - t^2\Phi_p(t) = \lambda\Phi_p(t), \tag{A.1}$$

where $\Phi_p(t)$ represents the Hermite functions or Hermite-Gaussian functions [54]. This equation can be expressed using operator notation for the left-hand side as

$$(\mathbf{T} + \mathbf{F}\mathbf{T}\mathbf{F}^{-1})\Phi_p(t) = \mathbf{S}\Phi_p(t) = \lambda\Phi_p(t), \tag{A.2}$$

where $\mathbf{T} = \mathrm{d}^2/\mathrm{d}t^2$ and $\mathbf{F}$ denote the differential operation and the standard FT, respectively. Here, we denote the operator $\mathbf{S} = \mathbf{T} + \mathbf{F}\mathbf{T}\mathbf{F}^{-1}$.

To incorporate the scaling transform, we replace t with t/σ in Eq. (A.1), leading to the following differential equation

$$\sigma^2 \cdot \frac{\mathrm{d}^2}{\mathrm{d}t^2}\Phi_p\left(\frac{t}{\sigma}\right) - \left(\frac{t}{\sigma}\right)^2 \cdot \Phi_p\left(\frac{t}{\sigma}\right) = \lambda\Phi_p\left(\frac{t}{\sigma}\right). \tag{A.3}$$

Similar to Eq. (A.2), this can also be rewritten using operators $\mathbf{T}$ and $\mathbf{F}$ as

$$(\mathbf{T} + \sigma^4\mathbf{F}\mathbf{T}\mathbf{F}^{-1})\Phi_p\left(\frac{t}{\sigma}\right) = -\sigma^2\lambda\Phi_p\left(\frac{t}{\sigma}\right). \tag{A.4}$$

This equation gives rise to a "n^2 matrix method" [55], corresponding to the scaling stage $\mathbf{H}_\sigma$ matrix in **Section** 2.1.

Given that the operator $\mathbf{S}$ can be expressed as $\mathbf{S} = \mathbf{T} + \mathbf{F}\mathbf{T}\mathbf{F}^{-1}$, where $\mathbf{T}$ is a second-order difference operator represented by the circulant matrix corresponding to the impulse response $h[n] = \delta[n+1] - 2\delta[n] + \delta[n-1]$, and $\mathbf{F}\mathbf{T}\mathbf{F}^{-1}$ is a diagonal matrix, we can replace the n^2 matrix with the $\mathbf{S}$ matrix, defined as $\mathbf{S}_\sigma = \mathbf{T} + \sigma^4\mathbf{F}\mathbf{T}\mathbf{F}^{-1}$ [47]. This approach mirrors the method defined for DFRFT in reference [36], where a matrix $\mathbf{P}$ is introduced to decompose any vector $x[n]$ into its even and odd components. The $\mathbf{P}$ matrix maps the even parts of an N-dimensional vector $x[n]$ to the first $\lfloor (T/2+1) \rfloor$ components and maps the odd parts to the remaining components.

Considering a method similar to the eigendecomposition of the n^2 matrix, we perform an eigendecomposition on the transformed matrix $\mathbf{PS}_\sigma\mathbf{P}^{-1}$ and obtain a block diagonal matrix

$$\mathbf{PS}_\sigma\mathbf{P}^{-1} = \mathbf{PS}_\sigma\mathbf{P} = \begin{bmatrix} \mathbf{Ev} & 0 \\ 0 & \mathbf{Od} \end{bmatrix}, \tag{A.5}$$

where the $\mathbf{P}$ matrix is both symmetric and unitary, i.e., $\mathbf{P} = \mathbf{P}^{\mathrm{H}}$. The eigenvectors of $\mathbf{PS}_\sigma\mathbf{P}^{-1}$ can be determined separately from the $\mathbf{Ev}$ and $\mathbf{Od}$ matrices, while the corresponding eigenvectors of $\mathbf{S}_\sigma$ are merely the even/odd expansions of the $\mathbf{PS}_\sigma\mathbf{P}^{-1}$ eigenvectors. These eigenvectors correspond to the new scaling transform matrix $\mathbf{E}_\sigma$ obtained subsequent to the DFRFT.

In light of the new scaling transform defined on cyclic graphs, we can adapt the diagonal matrices associated with both the DFRFT and CM stages to be based on cyclic graphs. Specifically, the diagonal matrix for the DFRFT stage is given by $\mathbf{D}^\alpha = \mathrm{diag}\{\exp[-i\pi m\alpha/2]\}$, and for the CM stage by $\mathbf{D}^\xi = \mathrm{diag}\{\exp[i\pi\xi(n-\frac{N}{2})^2/N]\}, 0 \le n, m \le T-1$.

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